

Quantum Computing Algorithms and Results for H atom and 2-qubit Ising Model

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1 Introduction

Quantum computing enables simulation of quantum systems, solving eigenvalue problems, and studying models of interest for condensed matter and quantum chemistry. This report details solutions and results from a notebook exploring ground-state calculations for a hydrogen atom and a model quantum Ising Hamiltonian, using variational algorithms executed via Qiskit.

2 Methods

2.1 Quantum Algorithms and Circuit Ansatz

Calculations employ the Variational Quantum Eigensolver (VQE) algorithm with hardware-efficient TwoLocal ansatz and classical optimizers. Hamiltonians are derived using Qiskit Nature for molecular systems or custom code for the Ising model, and are mapped to qubit operators (Pauli strings) via the Jordan-Wigner transformation. All calculations are performed in a noiseless simulator using Qiskit Aer.

2.2 Problems Addressed

1. Single Hydrogen Atom (H)
2. Custom Two-Qubit Ising Model Hamiltonian

2.3 Optimization and Simulation

Optimization is performed using L-BFGS-B with a maximum of 100 iterations. VQE converges the ground state energy of the given Hamiltonian; exact energies are also obtained via classical diagonalization (NumPyEigensolver) for comparison.

3 Problem 1: Ground State of Single Hydrogen Atom

System: H at (0, 0, 0), Charge 0, Spin 1, Basis STO-3G.

Hamiltonian generated using PySCFDriver. Mapped to 2 qubits.

Ansatz: TwoLocal (linear entanglement, 1 repetition, 'ry' and 'rz' rotation blocks)

Results:(Rounded)

- VQE energy (TwoLocal): -0.46658183 Ha
- Exact energy: -0.46658185 Ha
- Error: 2.30×10^{-8} Ha (1.44×10^{-5} kcal/mol)
- Nuclear repulsion energy: 0.0
- Qubits used: 2

Analysis: The TwoLocal ansatz demonstrates extremely high accuracy relative to the exact diagonalization for the hydrogen atom.

4 Problem 2: Ising Model Hamiltonian

System: Two-qubit custom Ising Hamiltonian

Hamiltonian: ZZ nearest-neighbor interaction, constructed as `SparsePauliOp(['ZZ'], coeffs = [1])`

Ansatz: TwoLocal (full entanglement, 2 repetitions)

Results:(Rounded)

- VQE energy (TwoLocal): -0.99999999997
- Exact energy: -1.0
- Error: 2.85×10^{-11} (2.0×10^{-8} kcal/mol)
- Qubits used: 2

Analysis: The TwoLocal circuit with proper configuration can discover the exact ground state for simple Ising models.

5 Methodological Features and Functions

- **Jordan-Wigner mapping** used for all qubit encodings.
- **Ideal Estimator:** Simulations are noiseless.
- **Optimizer:** L-BFGS-B with callback for convergence monitoring.
- **Comparison:** All energies are compared to exact diagonalization.

The `ising_hamiltonian` function enables easy construction of Ising models with arbitrary qubit count and interaction strength. Commented lines suggest future extension to field terms and anisotropic models.

6 Limitations

All calculations are performed without noise or decoherence. Real hardware may result in significantly reduced accuracy. Studied systems are extremely small; methods illustrated are scalable with quantum hardware for larger models or many-body systems. Restriction to TwoLocal ansatz may limit flexibility in discovering more complex ground states.

7 Conclusion

This notebook showcases VQE for molecular and spin Hamiltonians. Near-exact accuracy for small systems is demonstrated, providing a foundation for larger, more challenging systems as quantum hardware matures.

8 References

- Qiskit Documentation: <https://qiskit.org/documentation/>
- Qiskit Nature: <https://qiskit.org/ecosystem/nature/>
- S. McArdle, S. Endo, A. Aspuru-Guzik, et al. "Quantum computational chemistry." Reviews of Modern Physics **92**, 015003 (2020).
- VQE and TwoLocal Ansatz: Peruzzo et al, Nat. Comm. 5, 4213 (2014).